A Federated Digital Twin Framework with Multimodal Data Integration for Personalized Healthcare Monitoring.

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Capstone Thesis (CSE 400) submitted in partial fulfillment of the requirements for the degree of

Bachelor of Science in Computer Science and Engineering

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I declare that this thesis entitled *A Federated Digital Twin Framework with Multimodal Data Inte- gration for Personalized Healthcare Monitoring* is the result of my own work except as cited in the references. The thesis has not been accepted for any degree and is not concurrently submitted in candidature of any other degree.

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Dedicated to my beloved parents
for their unconditional love and
support, and to everyone who
inspired, guided, and encouraged
me throughout my journey.

– Sourav Debnath

Dedicated to Almighty
Allah —
*All praise to Him for His
endless mercy and guidance.
Dedicated to my beloved
parents for their love and
support, and to all who
inspired and encouraged me
along the way.*

– Nur Hasan Hasib

To my ever-loving and
supportive parents,
*whose sacrifices, prayers, and
unconditional love have been
my greatest strength.*

– Sheak Fahad

ABSTRACT

In the time of fast growing , the health technologies are also growing rapidly. The need for personalized and continuous health monitoring has increased a lot. However, traditional centralized healthcare systems struggle to combine data from different sources while also keeping patient information private and secure. Centralized architectures are vulnerable to data breaches and inefficient in managing multimodal data generated from wearable devices, electronic health records, and environmental sensors. This work proposes a federated digital twin–based framework for secure, privacy-preserving, and personalized healthcare monitoring. There are many existence healthcare systems but they faces different type of problem. Like scalability and store data in the system. Which makes more difficult to cooperation. So, overcome these issues,we proposed a framework that uses digital twin technology that create virtual models of individuals persons. These models are dynamic and continuously update by the changes of patient health. Federated learning is used to train models on edge devices without any sharing raw data that helps to protect privacy and follow regulations. Combining both of them and used it for collect data from different resources and combined the attention based methods to support the real time prediction and also early detection for abnormal health conditions at the edge level. The experimental results shows that the proposed models achieve highly accurate and reliable patient status prediction, with the ensemble MXBoost model delivering the best overall performance 99.72 percent accuracy and near-perfect AUC. Overall, this research offers a scalable, secure and efficient solution for personalized healthcare monitoring and lays a strong foundation for future real world use.

Keywords: Digital Twin, Federated Learning, Personalized Healthcare, Multimodal Data Fusion, Privacy Preservation, General Data Protection Regulation,The Health Insurance Portability and Accountability Act

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1. Introduction

1.1 Background of the Study

Healthcare is changing a lot these days. It's moving from just treating problems when they show up to something more ongoing and tailored to each person. All of this new gadgets like wearable ans sensors connected through the internet of things, doctors can now real time info how the patient lead their life. So, this data is very useful when combined data with this technologies. Like Artificial Intelligence and computed on edge level or in the cloud. Together they can turn raw data into any type of real support of health care. Such as real time monitoring and adjust treatment whenever needed. Digital twins also play an important role, because this create a virtual version of real patient with their data. Models also helps out how the body works and possible treatment. Early studies also this digital twin as a tracker of body.
In this research we also learn that health data is more complex because it comes from different sources. It also includes medical records from doctor visits, genetic info, time based sensor data and also details about the person lifestyle. And all of these data bringing in a single digital twin is not so easy.It also required proper methods to combine the data and clean it and also make sure everything matches properly. Moreover, using digital twins in real life hospital takes several challenges. And data privacy is major concern, along with systems compatibility and the ability to add more patient.Generally hospitals don't share patient raw data to anyone due to security risks and strong regulations such as HIPAA and GDPR place further limits on how data can handled. So, as a result, achieving real time analysis under this restriction requires systems that both are efficient and accurate. And these all challenges motivate the need to expore new system architectures that combine digital twins with federated learning and privacy preserving techniques.[1]

1.2 Problem Statement

Already digital twins have shown a very good results in early research.But there is still a major problem with that and stop it from being widely used in healthcare.So, this problems make it harder to apply digital twin and federated larning systems in real hospitals and clinics.

So,before dive into the problems discussion.It is very important to understand that most current digital twin solutions work well in small or controlled research settings. But when it used in large scale or real healthcare systems.They often fails.Many of them do not provide proper personalization for individuals.And also can not properly combined data from different sources.And also do not concern about privacy and rules regulations about data protections.Most existing systems do not scale well beyond small test projects.So, this create a gap between research ideas and real clinic or hospitals use.After all of these reasons, there is a strong need for a new framework that combines digital twins,federated learning and multi-modal data integration. That support secure,scalable and personalized healthcare.

A. **Insufficient personalization:** Most existing digital twins systems are made by general models based on large scale.But these models do not fully capture the unique genetic background,lifestyle and environment of every patient.As a result the predictions are not accurate and it unable to prove personalized treatment or health advice properly.[1]

B. **Fragmented multi-modal data:** Patient data comes from different sources such as electric health records, hospitals records,wearable devices and different software platforms and genetic databases.So this data is stored in different formats and systems.So all of these different data type are difficult to combining with digital twins.So many of current digital twins do not have effective methods to clean ,align and merge this data properly.[2]

C. **Privacy and data sharing issues:** Existing centralized data collection methods can increase the risk of data leaks and security problems.So this makes it hard to follow the strict privacy laws such as HIPAA and GDPR. Causes of these risks,hospitals and healthcare organizations are not willing to share the patient data with other systems.So as a result it slow downs the real world use of digital twins.[3]

- **D. Limited scalability and interoperability:** From existing digital twin systems are tested on small experiments.And can not easily expand to support many hospitals or large number of patients.Moreover they also not working properly with common healthcare standards such as HL7 and FHIR. So, which encourage us to real clinical deployment.[4]

1.3 Research Gap

The latest improvements in digital twins, federated learning and mutlimodal data fusion have helped to reduce some of the early problems.But, many of the still remaining.If we look closely at current research.It becomes clear that most progress has been made only in theory or small experiments.But Fully completed and practical systems that combine all of these technologies at once still limited.So focusing to solve this problem.And build digital twin systems that are not only advanced in design but also practical and suable in real life scenario.

- **A. Limitated use of federated digital twins:** There is a successful use of federated learning in areas such as medical imaging and electronic record analysis.But basically we use for text and numerous data.however, use it in patient level digital twin system still rare.There is only few studies that show how digital twins can updated continuously using different type of data in a federated way.So this leaves a large gap and an open area for furthure research.[5]
- **B. Weak combination of different data fusion:** Exist digital twin often focus only on basic health signals, such as heart rate or activity data They often ignore other major information, like genetic data,lifestyle habit,environmental factors.So because of that, current digital twins unable to fully responsible.[6]
- **C. Gaps in privacy and governance frameworks:** Federated learning supports ideas like secure data sharing and encryption.But stills there are many existing solutions do not properly address the privacy laws,governance models or systems standards such as HL7 and FHIR.So which are required for real life use.[7]

1.4 Motivation

In this context our main motivation is to design and evaluate a **Federated Digital Twin framework** that uses data from multiple sources to support personalized health monitoring.And our proposed system aims to solve the several problems that found in the current approaches. Most recent one are struggle to handle data coming from different sources and often fail to properly protect patient privacy on a large scale. As a result it do not provide enough personalization.

To addressing this problem our framework combines federated learning with multi modal data integration.And digital twins allows to build individuals virtual patients and continuously updated it as new data available. And key features of this system is that patient data remains in local devices or within the hospital systems.And this data never send to the central server. So this helps to protect patient privacy and ensure the rules. Also this framework uses both edge devices and cloud computing. For that it allows real time patient monitoring with minimal delay. It also works with existing hospital infrastructure.

Another motivation of this research to show how modern AI models can trained together. In a federated environment to make prediction without let data into any type of risks. There is some specific motivation of this research:

A. **Architecture design:** Create a innovative design for the federated digital twin system.This will show data collection,cleaning and how that is used for model train locally.It also have secure ways to combine updates from different devices and manage them.

B. **Multi-modal fusion pipeline:** Build strong methods to combine different type of data,like time based health signals,hospital records,genetic or lifestyle information. This pipeline will handle data type,missing information and varying collection rate.To make sure each patient digital twin is accurate or not.

C. **Privacy-preserving training:** Federated learning use for encryption and secure aggregation to protect the patient privacy.Purpose is to show the system can create the global models accurately without sharing raw data.

D. **Real-world evaluation:** Test the frameworks on realistic datasets.And measures how well it works in terms of accuracy,speed,reliability and privacy protection.Also include comparing the federated system to the traditional centralized digital twin models.So we can see the difference above them.

E. **Interoperability and deployment design:** Show the ways to connect the system with real hospital system following the standards.This ensure that not only theoretical but also ready to used in actual life.

1.5 Research Objectives

To achieve the motivation mentioned above, this research has some objectives. Each use for build a complete and usable federated digital twin system.

A. To Designing a federated digital twin system that can combine different type data such as health signals,genetic information and life style details.

B. To develop methods that can protect patient privacy while allowing the system to learn from data stored in different hospitals and devices without sharing any raw data.

C. To create a hybrid system using edge devices and cloud computing,so that patient can monitor them self in real time.

D. To make sure the system works with the healthcare standards like HL7 and FHIR,so it can used in real hospitals and clinics.

E. To provide both ideas and practical steps that help digital twins become useful in real world.

1.6 Research Contributions

The contributions of this thesis are both technical and practical,aim to improve digital twin research and giving them to useful guidance for healthcare use.Present digital twin focused only single type data and centralized training.But this works present a**federated,multi modal and privacy focused design**.We are not eagerly just to make accurate models also follow the strict rules and safety.The contributions include new federated digital twin architectures,methods combining in different data types,maintaining the privacy and through testing show the framework results in practical life.

- **Federated Digital Twin Architecture:** A system that combines edge and cloud computing with federated learning.And this allows real time monitoring without sharing any raw data.
- **Multi modal Fusion Pipeline:** Through this pipeline methods and tools for combining different types of healthcare data into single or unified view of each patient.
- **Privacy-first Training Strategy:** Federated learning used for maintain the privacy and follow the rules with the encryption and secure data aggregation.
- **Experimental Evaluation:** Test showing that federated digital twins can perform well in practical life.And also in large scale of patient and large areas.

1.7 Financial Planning and Timeline

In financial planning and timeline it represent the costs and planned timeline of this research works.Its purpose to show the proposed thesis is practical and feasible in both term budget and management.

1.7.1 Budget Estimation

Table 1.1: Resource Allocation and Budget Overview

Resource Item	**Source / Specification**	**Estimated Cost**
Development Tools	Scikit-learn, MXBoost,Tensorflow, Py-torch	Free/Open-source
Dataset Access	Human Vital Signs Dataset	Public Domain
Computation	Google Colab,Kaggle environment	Free
Hardware Support	Personal Laptop and PC	70,000tk
Net Expenses		$70,000tk

Table 1.1 presents the financial statement that no major cost in this research. Because major things are mostly free and open sources. This table shows that resource item and source and also estimated cost above all things.

1.7.2 Gantt Chart

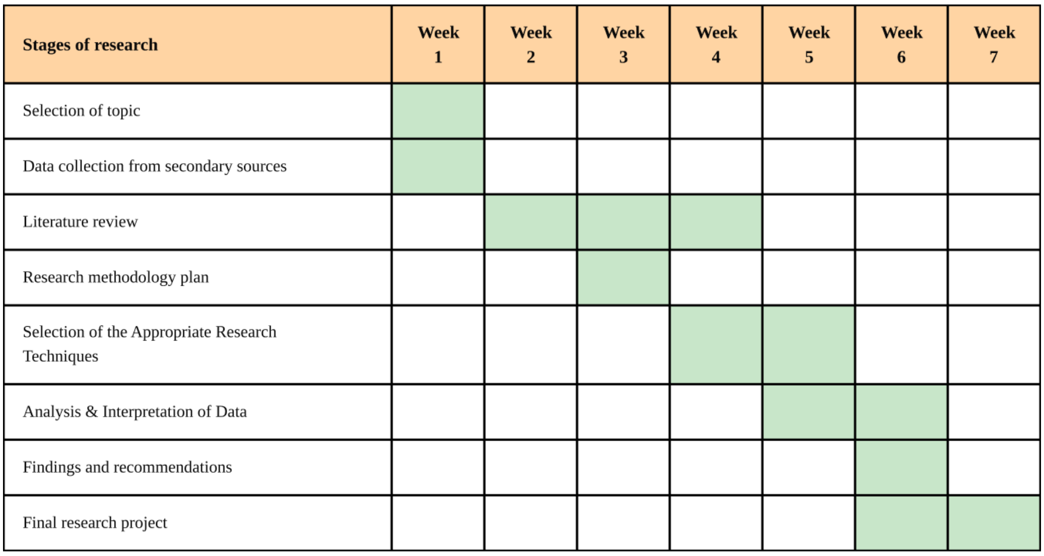


Figure 1.1: Proposed Gantt Chart for Research Timeline

Figure 1.1 shows the planned of this research timeline. Easy visibility we use a gantt chart. This chart preents the main research activites week by week. And following that easier to follow the project in an organized way.
This the started with choosing the thesis topic and followed by collecting the data from secondary and public sources.Then a detailed literature review was done for that study with existing works.Based on literature a complete research methodology was created that include some machine learning models and designing the system architecture. Later stages of the research focused on analyzing the data and test the model and interprets the result.Lastly, the study wrapped up with drawing conclusions, and giving recommendations and preparing thesis report.
The gantt chart helps to show the tasks are interconnected together,manange the time effectively and tracking the progress.

1.8 Organization of the Dissertation

This is dissertation basically about the several chapters and their structure:
Chapter 1, **Introduction** this research study presents its research background together with its problem statement and research gap and study goals and study findings. The study describes its motivation to create a federated digital twin framework which will enable personalized healthcare monitoring.
Chapter 2, **Literature Review** previous research about digital twin technology federated learning multi-modal data integration methods and privacy-protecting healthcare systems. The study demonstrates current method limitations while providing foundational support for the proposed federation-based digital twin system.
Chapter 3, **Proposed Methodology**, describes the overall system architecture of the federated digital twin framework. It explains the multimodal data acquisition process, edge-level digital twin construction, federated learning strategy, blockchain-based verification mechanism, and personalization of the global digital twin. Mathematical formulations, data flow diagrams, and algorithmic steps are also presented in this chapter.
Chapter 4, **Data Presentation and Analysis**, discusses the datasets used in this research, data preprocessing techniques, feature categorization, and statistical analysis. It provides a clear understanding of the multimodal data structure and its role in digital twin modeling.
Chapter 5, **Engineering Considerations**, examines the societal, environmental, and ethical aspects of the proposed system. This chapter highlights the broader impact of the research and ensures alignment with sustainable and responsible engineering practices.
Chapter 6, **Result Analysis and Discussion**, presents the experimental results, performance evaluation of different machine learning models, confusion matrix analysis, ROC curves, and digital twin simulation outcomes. A comparison with existing literature is also included to validate the effectiveness of the proposed approach.
Chapter 7, **Conclusion and Future Works**, summarizes the overall findings of the research, discusses its limitations, and provides directions for future research and real-world deployment of federated digital twin systems in healthcare.

2. Literature Review

2.1 Overview

Systematic and comprehensive view of the literature concerning the Digital Twin-based healthcare systems, Federated Learning for privacy-preserving model training, and multimodal data integration for personalized health monitoring. Among the reviewed works, there are baseline studies, the latest journal articles, and graduate theses that broadly look into the issues of architectural design, paradigm of learning, and security mechanisms for distributed healthcare analytics. The digital twin's changing nature by making use of diverse providers' medical data, the federated learning allowing collaboration without sharing data, and the blockchain as a privacy and trust provider are the main topics of discussion. While the studies indicate the possible and practical applications of systems with federations and multimodal digital twins, they also highlight the still-existing difficulties regarding scalability, real-time deployment, dummy multimodal fusion, and large-scale clinical validation. The findings and deficiencies pointed out in this chapter provide the basis for the federated digital twin framework that will be proposed later in this thesis.

2.2 Related Works

This review of the literature discusses the most significant research work and academic theses that form the foundation of this study titled **A Federated Digital Twin Framework with Multimodal Data Integration for Personalized Healthcare Monitoring**. Out of these, seven are directly used as base papers that contribute conceptually and methodologically to this research, while the remaining three are recent conference and journal papers that provide supporting context. These works collectively address major themes such as digital twins (DT) in healthcare, federated learning (FL) for distributed model training, multimodal data fusion, privacy, and interoperability. Together they establish the scholarly base upon which this thesis builds its proposed federated multimodal DT framework.

A comprehensive review published a long time ago as well as the use of Smith et al. [1] suggested the Digital Twin integrated with Federated Learning for a secure multi- institution model training. However, they further pointed out that the research only allowed Prototype-level validation to be done and gave limited multi modal coverage.

Lee et al. [8] did a great job in reviewing all the federated learning methods used for multimodal healthcare data and covering them thoroughly. The paper provides not only the conceptual insights for major data fusion but also the lack of implementation and experimental validation.

According to Garcia et al. [9], the digital twins powered by AI were analyzed for personalized healthcare monitoring, where predictive analytics and real-time decision-making were pointed out as their main features. Though it is a strong conceptual contribution, the research does not cover federated or decentralized learning approaches.

A hybrid system that includes blockchain, digital twins, and federated learning was proposed by Patel et al. [10] as a way to improve healthcare security. The solution increases privacy and trust but primarily addresses security issues and is thus not very widely used in the real world yet.

In the paper by Zhang et al. [11], global learning digital twins were suggested for personalized medicine which, in fact, do not stop to adapt with new incoming patient data. The proposed method allows for the personalization of prediction, but still large-scale validation is hindered by the limited dataset scope.

Kumar et al. [12], on the other hand, came up with a multimodal biomarkers fusion approach that is incorporated in digital twin systems for chronic disease management. The provided evidence of improved accuracy in monitoring, however, its use is restricted to certain disease conditions only.

Rahman et al. [13] coupled federated learning and digital twins to achieve privacy-preserving prediction of cardiovascular diseases. The research shows that it is possible to collaborate between hospitals across different locations, but still, it is limited to a single disease and lacks generalised multimodal extensibility.

Lauer-Schmaltz [14] built patient-specific digital twins by integrating multimodal data streams that included electronic health records (EHRs), sensors, and imaging. Although the authors share some practical considerations regarding the implementation, their framework does not take into account federated or distributed learning architectures.
Adegbola [15] investigated the combination of blockchain and federated learning for attaining security in distant healthcare systems. The research focuses on the decentralization of privacy preservation, but its evaluation is restricted to simulation and does not involve multimodal data integration.
Chen et al. [16] introduced a privacy-preserving federated cognitive digital twin framework by means of encrypted model aggregation. The method guarantees both confidentiality as well as accuracy; however, it has been tested either on synthetic datasets or datasets with limited access, and no large-scale clinical validation has been carried out.

Table 2.1: Comparison of Related Works

Author(s) / Year	**Key Contribution**	**Challenges Identified**	**Limitations / Gaps**
Smith et al. (2024) [1]	Proposes Digital Twin integration with Federated Learning for secure multi-institution model training.	Cross-site data heterogeneity and secure synchronization.	Prototype-level validation only; limited multimodal coverage.
Lee et al. (2024) [8]	Reviews FL with multi-modal healthcare data integration.	Synchronization and cross-modal imbalance.	Lacks implementation; conceptual survey work.
Garcia et al. (2025) [9]	Analyzes AI-driven DTs for personalized healthcare monitoring.	Integration of streaming sensor data with static clinical records.	Does not address federated or decentralized learning modalities.
Patel et al. (2025) [10]	Combines Blockchain, DT and Federated Learning for secure healthcare monitoring.	Blockchain latency and model convergence delays.	Focused on security layer; limited real-world deployment.
Zhang et al. (2025) [11]	Proposes global learning digital twins for personalized medicine.	Real-time adaptation and patient specific calibration.	Large scale tested on limited dataset scope.
Kumar et al. (2025) [12]	Fuses multimodal biomarkers in digital twin systems for chronic disease.	Temporal alignment, missing features, feature sparsity.	Application specific; lacks generalization across disease types.
Rahman et al. (2024) [13]	Integrates Federated Learning and Digital Twins for cardiovascular prediction.	Privacy-preserving distributed learning under heterogeneous hospital data.	Single disease focus; not generalized multimodal approach.
Lauer-Schmaltz (2024) [14]	Implements patient-specific DT using multi-modal data streams (EHR, sensors, imaging).	Data preprocessing heterogeneity and model alignment.	No federated learning or distributed architecture considered.
Adegbola (2024) [15]	Applies Blockchain and Federated Learning for remote healthcare systems.	Trade-offs between decentralization and latency, computational cost.	Simulation design only; lacks multimodal data integration.
Chen et al. (2025) [16]	Proposes privacy-preserving federated cognitive digital twin framework.	Encryption cost, scalability of aggregation and latency.	Tested on synthetic or limited data; no broad clinical trial.

Table 2.1 presents a comparative summary of recent studies on Digital Twins, Federated Learning, and multi-modal healthcare systems. The table highlights the key contributions of each work, the challenges they address, and the remaining limitations. This comparison makes it easier to understand current research trends and clearly identify existing gaps that motivate the proposed research.

2.3 Summary

This research is about the digital tiwns in healthcare systems. Also how they related to the federated learning, digital twins and different types of data and also security preserving. Digital twins looks pretty useful for mapping out how a specific patient is doing physiologically. Federated learning lets places like hospitals train models together without actually swapping patient data around, which is a big deal for privacy. Some papers talk up blockchain too, saying it builds trust and keeps data secure in these scattered setups. But honestly, a lot of this work treats each part separately, not really combining them into one solid system, which feels like a missed chance. The challenges come in the literature. Dealing with all sorts of data types that don't match up easily, making sure things adapt in real time, handling the extra communication and computing load, and scaling it for actual clinics. Plus, it seems like most are just prototypes or tested in simulations, or focused on one disease, without broader checks in real hospitals or ways to extend to more data kinds. That part gets a bit messy, I think, because without those tests, its hard to know how far it goes. Overall, this points to needing a better framework that pulls federated digital twins together, keeps privacy intact, scales up, and mixes multi-modal data for ongoing patient watch. It motivates what comes next in the thesis, the proposed setup in later chapters, though we are not totally sure how seamless that integration will be yet.

3. Proposed Methodology

3.1 Overview

The proposed methodology presents a complete and privacy focused **Federated Digital Twin** framework.That use for continuous,intellectual and personalized monitoring.It brings the major three advanced technologies together-**Digital Twins,Federated Learning and Multi modal**.That provide secure ,adaptive and understandable healthcare across the medical environment.
Unlike The traditional centralized healthcare systems, this proposed model keeps all sensitive data within each local hospital or clinic.While using the combined knowledge from all participants to global one.Federated learningsteps and secure combination of model updates improve the overall diagnostic system without putting the patient data into any risks.
The modular architecture, establishes a workflow that interlinks IoT-based patient sensing, edge-level digital twin construction, global federated aggregation, blockchain validation, and personalized model adaptation.

The methodology proceeds through five principal stages:

A. Multimodal data acquisition and preprocessing.

B. Edge-level digital twin construction and local training.

C. Federated learning and model aggregation.

D. Cryptographic-based verification and ledger maintenance.

E. Global digital twin update and personalization.

Each stage is described in detail below, accompanied by mathematical formulations and algorithmic steps.
In this research we introduced an enhanced proposed methodology diagram that supports various features like privacy consideration,digital twin creations,federated learning integration and so on. So below we design an proposed methodology diagram based on our research.

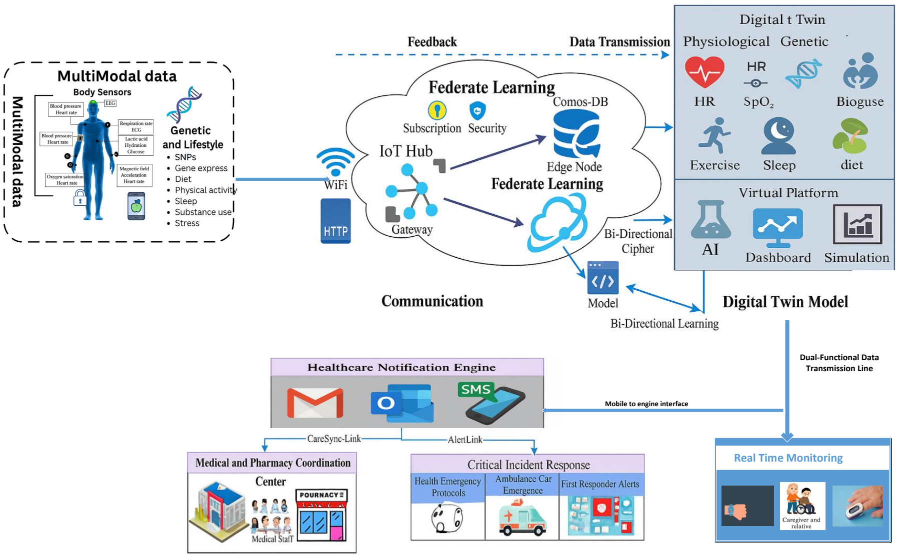


Figure 3.1: Proposed architecture of the federated digital twin framework

Figure 3.1 presents the overall architecture of the proposed federated digital twin framework. The proposed system is described in detail in the following sections with supporting mathematical and algorithmic steps. This research introduces an enhanced methodology that integrates privacy protection, digital twin construction, and federated learning. Figure 3.1 presents the overall architecture of the proposed federated digital twin framework.

3.2 Multimodal Data Acquisition and Preprocessing

The first stage focuses on acquiring heterogeneous medical data streams originating from wearable IoT devices, clinical records, and lifestyle-tracking applications. These modalities form the foundational layer of each patient's digital representation. Each patient p produces multimodal data:

$$X_p = \{x^{(1)}, x^{(2)}, \ldots, x^{(M)}\} \tag{3.1}$$

where $x^{(i)}$ represents the i^{th} data modality:

- **Physiological data:** Continuous signals such as heart rate (HR), oxygen saturation (SpO_2), blood pressure (BP), and body temperature (BT).
- **EHR text:** Clinical notes, prescriptions, diagnostic codes, and lab results.
- **Lifestyle metrics:** Step count, calorie eaten, and stress levels.

All types of data is first processed separately to make it consistent and reliable. Through the pre processing missing values are filled in, unneccessary data are removed and data convert in a standard range.

$$z^{(i)} = f_{\theta_i}(x^{(i)}), \tag{3.2}$$

here, f_{θ_i} represents the encoder or features extraction model used for data type i. All transformed features are then combined into a single and unified representation:

$$z_p = \text{Concat}\big(z^{(1)}, z^{(2)}, \ldots, z^{(M)}\big) \tag{3.3}$$

this forming combined multi modal representation of patient p.
For time-based signals, noise reduction is applied using rolling averages:

$$\overline{x}_w(t) = \frac{1}{w} \sum_{i=0}^{w-1} x(t-i) \tag{3.4}$$

where w is the sliding window width.

3.3 Edge-Level Digital Twin Construction and Local Training

At this point, each healthcare institution or IoT gateway functions as a **local node**, it creates a separated **Digital Twin (DT)** for each patients. The Digital Twin models hold the patient's physical health and learns predictions patterns using only the data is available in local site.
For node i, the DT model is denoted by:

$\mathrm{DT}_i = \mathcal{M}_{\phi_i}(z_i)$(3.5) where $\mathcal{M}_{\phi_i}$ is the local model with parameters ϕ_i. The node minimizes:

$$\mathcal{L}_i = \frac{1}{n_i}\sum_{j=1}^{n_i} \ell(\mathcal{M}_{\phi_i}(z_{ij}), y_{ij}), \tag{3.6}$$

using stochastic gradient descent:

$$\phi_i^{t+1} = \phi_i^t - \eta \nabla_{\phi_i} \mathcal{L}_i. \tag{3.7}$$

Each model uses the combine model that include MXboost,Multi layer perceptron and XGboost for effective performance:

$$P_{\text{final}} = \text{Mode}(P_{\text{MLP}}, P_{\text{XGB}}). \tag{3.8}$$

To handle the different types of data and the information is grouped based on feature patterns before training. This approved seperate model paths for numerical, catagorical and time based data.

3.4 Federated Learning and Model Aggregation

After local training is finished each institution send only ecrypted model paramters to central.No of the raw patient share there data.The global model is then created by taking the weighted average of all locals:

$$\phi_{\text{global}} = \sum_{i=1}^{N} \frac{n_i}{\sum_{j=1}^{N} n_j} \phi_i. \tag{3.9}$$

One round is completed the updated global model send back thedata to all locals for the next round:

$$\phi_i^{(t+1)} \leftarrow \phi_{\text{global}}^{(t)}. \tag{3.10}$$

Make the process faster and reduce the cost taking an action that grouped the similar data patterns.This similarity measued using this formula:

$$\text{Sim}(DT_i, DT_j) = \frac{\langle z_i, z_j \rangle}{\|z_i\| \, \|z_j\|}. \tag{3.11}$$

The system reduces unneccessary data exchange and helps to the model converge more quickly

3.5 Cryptographical-Based Verification and Ledger Maintenance

Each federated update is saved as a secure transaction that can be verified later:

$$T_i = \langle \text{NodeID}_i, H(\phi_i), \text{Timestamp}, \text{Signature} \rangle. \tag{3.12}$$

This record stored the node identity.Also include a hash of the model update ,time and digital signature submitted.Also use a protocol to maintain the network. This ensures every update is validated before being accepted globally.

Only hashed metadata are recorded on-chain, while encrypted model weights are stored off-chain for privacy. Encryption and role-based access control are formalized as:

$$E_K(D) \to C, \qquad H(D) \to H', \qquad AC(U, R) \to P. \tag{3.13}$$

The cryptographic thus guarantees immutability, accountability, and traceability of the training process.

3.6 Global Digital Twin Update and Personalization

After validation, each node personalizes the global model using its latest patient data:

$$\phi_i' = \phi_{\text{global}} - \eta \nabla_\phi \mathcal{L}_i'. \tag{3.14}$$

This fine-tuned model becomes the patient's personalized digital twin, continuously synchronized with real-time physiological streams.
Anomalies are detected through deviation thresholds:

$$\text{Anomaly} = \{ 1 \text{ , if } \|P - P_{\text{historical}}\| > \theta, 0, \text{otherwise.} \tag{3.15}$$

3.7 Mathematical Summary of the Workflow

The complete iterative operation of the Federated Digital Twin system is described by:

Local Training: $\phi_i^{(t+1)} = \phi_i^{(t)} - \eta \nabla_{\phi_i} \mathcal{L}_i,$

Global Aggregation: $\phi_{\text{global}}^{(t+1)} = \sum_i \frac{n_i}{n_{\text{total}}} \phi_i^{(t+1)},$

Cryptographic Verification: $\text{Add}\Big(H(\phi_i^{(t+1)}),\ t\Big),$

Personalized Update: $\phi_i' = \phi_{\text{global}}^{(t+1)} - \eta \nabla_{\phi_i} \mathcal{L}_i'.$

These steps are repeated until convergence or until performance thresholds are achieved.

3.8 Data flow Diagram of Proposed Methodology

In this area we covered how can data proceed by any systems.However we analysis that how a data came from and how it going in different criteria and also manage their sustainability.Here, we design an data flow diagram that how a data flow under different situations and give an expected output.Below we give a figure of data flow diagram based on our proposed methodology.
Figure 3.2 shows the dataflow diagram based on the proposed methodology.In this section, we describe how data is processed within the proposed system. The analysis explains where the data originates, how it flows through different stages, and how it is managed to ensure sustainability. Based on this process, a dataflow diagram is designed to illustrate the movement of data under different operational scenarios and the expected outputs.

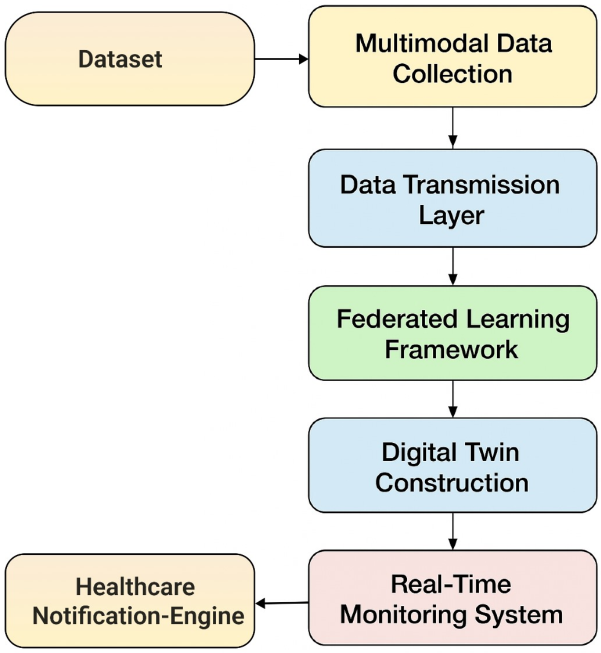


Figure 3.2: Dataflow diagram

3.9 Algorithms of Proposed Methodology

This section presents the core algorithms that operationalize the proposed Federated Digital Twin methodology. Each algorithm corresponds to a specific computational phase in the system pipeline—from multimodal data preprocessing to federated aggregation and personalized digital twin updates. All pseudocode has been designed for transparency, interpretability, and real-world implementability.

3.9.1 Algorithm 1: Multimodal Data Preprocessing and Feature Fusion

The first stage ensures the conversion of heterogeneous medical inputs (physiological, textual, and behavioral) into unified latent embeddings suitable for digital twin modeling. Each of this data has its own reprocessing routine before being fused into a joint multimodal representation.

Algorithm 1 Multimodal Data Preprocessing and Feature Fusion

Require: Raw data streams $\{x^{(1)}, x^{(2)}, ..., x^{(M)}\}$ for patient p
Ensure: Unified latent embedding z_p
1: **for** each modality $x^{(i)}$ **do**
2: Handle missing values and remove artifacts
3: Apply modality-specific normalization:
- Numerical: $x^{(i)} \leftarrow x^{(i)} - \mu\sigma$
- Textual: tokenization and embedding
- Time-series: smoothing via $\overline{x}_w(t)$

4: Extract latent feature: $z^{(i)} = f_{\theta_i}(x^{(i)})$
5: **end for**
6: Fuse embeddings: $z_p = \text{Concat}(z^{(1)}, z^{(2)}, ..., z^{(M)})$
7: **return** z_p

This algorithm combines data from sensor readings, EHR text, and lifestyle information. Each type of data is cleaned, normalized, and passed through an encoder network f_{θ_i} to create the basic feature vectors that collectively form the patient's multi modal digital profile.

3.9.2 Algorithm 2: Edge-Level Digital Twin Construction and Local Model Training

In this point, each healthcare institution creates its own edge digital twin model using local patient data. The MXBoost ensemble learns individualized patterns while maintaining local data effectively.

Algorithm 2 Edge-Level Digital Twin Construction and Local Model Training

Require: Preprocessed embeddings $\{z_p\}$ for local patients
Ensure: Trained local parameters ϕ_i
1: Initialize local model $\mathcal{M}_{\phi_i}$ (MXBoost: MLP + XGBoost ensemble)
2: Cluster embeddings by feature similarity
3: **for** each cluster C_k **do**
4: Compute loss $\mathcal{L}_i = \frac{1}{n_i} \sum_j \ell(\mathcal{M}_{\phi_i}(z_{ij}), y_{ij})$
5: Update parameters $\phi_i^{t+1} = \phi_i^t - \eta \nabla_{\phi_i} \mathcal{L}_i$
6: **end for**
7: Compute ensemble output $P_{\text{final}} = \text{Mode}(P_{\text{MLP}}, P_{\text{XGB}})$
8: Encrypt model weights $\text{Enc}(\phi_i)$
9: Transmit encrypted weights to federated aggregator
10: **return** $\text{Enc}(\phi_i)$

Each edge node trains its hybrid MXBoost model on local data, encrypts the parameters, and sends only encrypted weights to the central server. This ensures local privacy while still contributing to global intelligence.

3.9.3 Algorithm 3: Federated Aggregation and Cryptographic Verification

This algorithm controls how different nodes works together. Model updates are added into a global model only after crypto-based identity verification and hash checked, ensuring transparency and can not be denied of all learning contributions.

Algorithm 3 Federated Aggregation and Cryptographic Verification

Require: Encrypted local parameters $\{\text{Enc}(\phi_i)\}_{i=1}^N$
Ensure: Verified global model ϕ_{global}
1: **for** each node i **do**
2: Verify node identity and decrypt parameters $\phi_i = \text{Dec}(\text{Enc}(\phi_i))$
3: Compute model hash $h_i = H(\phi_i)$
4: Record transaction $T_i = \langle \text{NodeID}_i, h_i, \text{Timestamp} \rangle$ to Cryptographic
5: **end for**
6: Aggregate verified parameters:
$\phi_{\text{global}} = \sum_{i=1}^N n_i \sum_j n_j \phi_i$
7: Broadcast ϕ_{global} to all nodes
8: **return** ϕ_{global}

Before accepting any models updates, this checks each nodes using crypto.Only verified updates are added to the global model.Which keeps the models and systems secure.

3.9.4 Algorithm 4: Global Digital Twin Personalization and Anomaly Detection

This phase personalizes the federated global model for each individual patient. The digital twin continuously adapts to the patient's current data and monitors real-time changeable from baseline health patterns.

Algorithm 4 Global Digital Twin Personalization and Anomaly Detection

Require: Global model ϕ_{global}, local patient data (z_i, y_i)
Ensure: Personalized twin ϕ'_i and anomaly alert flag

```
Fine-tune global model: φ'_i = φ_global − η∇_φ L'_i
Predict current patient state: P = M_{φ'_i}(z_i)
Retrieve historical baseline: P_historical
if ||P − P_historical|| > θ then
    Trigger alert: Anomaly ← 1
else
    Anomaly ← 0
end if
Update blockchain record with new hash H(φ'_i)
return φ'_i, Anomaly
```

The model is adjusted using data from individuals patient so it can make personalized predictions accurately. An anomaly alert is triggered if the deviation exceeds threshold θ, and The updated model parameters are then securely recorded on the Cryptographic.

3.10 Summary

This research proposed the methodology is based on the designing a federated digital twin frameworks. The main idea of the methodology is to analyze health data without collecting all sensitive information in a centralized location. We generally focused on data are processed locally, and only necessary model updates are shared, which helps maintain data privacy. The methodology uses multi-modal health data, including physiological, behavioral, signal-based, and genomic features, to represent different aspects of an individual's health condition. Federated learning allows multiple local models to learn collaboratively, while a digital twin layer is used to create virtual representations of patient states. This makes it possible to monitor health changes and simulate different scenarios. Lastly, the proposed methodology provides a practical and secure framework for health data analysis. After all the system is evaluated in a simulated environment, it offers useful insight into how privacy-aware and personalized healthcare solutions can be developed in future applications.

4. Data Presentation and Analysis

4.1 Overview

The main things of this chapter is the data used in this research and its presentation and analysis. It explains the origin of the datasets, the reasons for selecting them, and the procedures followed to guarantee that the data is suitable for multi-modal health monitoring and digital twin–based experimenting. The chapter shows how healthcare data can be processed and arranged to meet the requirements of personalized and federated learning scenarios.

4.2 Data Collection

For this research work, data was mainly collected from the Human Vital Signs Dataset (2024). This dataset is publicly available on the Kaggle [17] platform and was created for general research use in healthcare and biomedical studies. The dataset was chosen because it contains different types of health related information, which is helpful when working with personal health monitoring and digital twin based ideas. In this study, the dataset was not treated as a complete clinical source, but rather as a base to test system behavior and data handling approaches.

Each record in the dataset represents a single individual. The data includes basic physical information such as age, height, weight and body mass index (BMI). Along with these, some common health measurements like blood pressure, heart rate, respiration rate and glucose level are also included. These values are commonly used in health monitoring systems and give a general idea about a person's physical condition. The dataset also contains ECG and EEG signal data, which are related to heart and brain activity. These signal based features were mainly used to increase variation in the dataset.

Apart from physical health data, some genomic related features are also present in the dataset. These include DNA fragments, SNP information and gene expression data. In this work, these features were not analyzed in deep biological detail. They were mainly kept to maintain the multimodal nature of the dataset and to observe how different types of data can exist together in one system. To increase the amount of usable data, another dataset with similar health and behavior related attributes was also added. This additional dataset was taken from the same platform so that the data structure and feature types remain compatible. The main purpose of adding this dataset was to reduce dependency on a single data source and to support broader experimentation.

During the merging process, similar features were aligned and duplicate records were removed. Numerical values were checked to avoid obvious mismatch or unrealistic values. This process was done carefully, although some minor inconsistencies may still exist due to the nature of preprocessed data.After combining the datasets, the final data included not only physical measurements but also lifestyle related information. These include dietary habits, physical activity level, sleep duration, smoking status and alcohol consumption. Such behavior related attributes are important because daily lifestyle can affect health condition over time.

4.3 Data Integration and Structure

After collecting the datasets, the next step was to integrate them into a single usable structure. This process was done step by step so that the data remain consistent and easy to work with. At first, both datasets were checked to see whether the attributes were similar and could be aligned without major changes. Only the attributes that matched in meaning and measurement were considered during integration . Common health related attributes such as height, weight, body mass index (BMI), blood pressure, heart rate, respiration rate and glucose level were matched between the datasets. ECG and EEG signal data were also included during this step. In most cases, the measurement units were already similar, so major conversion was not required. However, some small differences in naming and format were noticed.
Where attribute names or formats were different, simple standardization was applied. For example, similar attributes related to heart rate were renamed using a

single label so that confusion can be avoided later. Blood pressure values were kept in the usual systolic and diastolic format because this format is commonly used and easy to understand. Genomic related features such as DNA sequences and SNP information were also checked to remove repetition and to keep a similar structure across records.

4.4 Data Description

Dataset that used in this research is basically stored in a tabular form. Data structure, each row shows individuals patients data, and each column shows a specific type of data or measurement. This data format basically suitable for machine learning models. And it's easily to track and manage different types of information. Here, attributes in the dataset can be grouped into many categories based on their types.

A. Demographic Information:
Basic attributes such as patients ID, age, height, and weight . These attributes describe the basic physical characteristics of each participant. That are also used to observe general physical differences between people and to compute derived values such as body mass index .

B. Physiological Indicators:
Mainly this type of data are health-related measurements such as BMI, blood pressure, heart rate, respiration rate, and glucose level. These values represent core vital signs of the human body and are commonly used to assess cardiovascular, respiratory, and metabolic conditions.

C. Signal Data:
Signals based data such as ECG and EEG signals, these all type of data are electrical activity of the heart and brain. This are related to the physiological behavior. The signals are mainly used for anomaly detection.

D. Genomic and Molecular Attributes:
The dataset also contains some genomic and molecular level attributes. These include DNA sequence information, SNP variation and gene expression related data. These columns store genetic related information for each participant. In this study, these features were not used for deep genetic or medical

diagnosis. They were mainly included to understand how genetic data can be combined with other health data in a single system.

E. Behavioral and Lifestyle Attributes:
In addition to biological data, the dataset includes several behavioral and lifestyle related attributes. These include diet type, exercise level, sleep duration, smoking status and alcohol consumption. These attributes describe daily habits of individuals and provide additional context to health related measurements. Behavioral attributes are mostly qualitative in nature. They do not directly measure physical health, but they strongly influence long-term health outcomes. In this research, these features were included to study how lifestyle patterns may correlate with physiological indicators. They were mainly used for contextual analysis rather than for strict numerical modeling.

To better understand the structure of the integrated dataset, a small sample excerpt is shown in Table 4.1. The table presents a few selected records with different types of attributes, including demographic, physiological, signal-based, genomic and lifestyle data.

Table 4.1: Sample excerpt from the integrated Human Vital Signs Dataset

ID	Age	Ht (cm)	Wt (kg)	BMI	BP	ECG	EEG	HR	RR	Glucose	DNA Seq.	SNP	Diet	Sleep
P1	64	181	57	19.5	102/62	1.13	0.12	81	18	135	ATCG...C	rs467	Fast-food	6
P4	58	180	90	24.5	112/74	0.93	1.89	97	20	90	ATCG...G	rs535	Balanced	9
P9	45	169	64	28.9	127/66	0.95	3.98	100	19	118	ATCG...A	rs786	High-protein	9
P12	45	162	55	26.6	118/83	1.05	1.55	92	18	115	ATCG...T	rs631	High-carb	4

Table 4.1 represents a few selected records with different types of attributes, including demographic, physiological, signal-based, genomic and lifestyle data.This sample representation shows that the dataset contains different types of data together in one structure. It includes numeric values, categorical attributes and sequence-based genetic data. Because of this mixture, the dataset can support different kinds of analysis. In this research, the representation mainly helped in testing multimodal data handling and digital twin based modeling approaches rather than performing detailed clinical evaluation.

4.5 Data Pre-processing

Before starting any analysis or model training, the dataset went through a data preprocessing stage. This basically used for data handling because data are come from different sources. So, preprocessing is very essential there.

4.5.1 Data Cleaning

In data cleaning process, missing value were first identify and used pruned for missing value. Missing value are commonly in measurements type data and signals based attributes.So, there is some techniques like imputations was applied there. As a results, most frequent value was replace by missing entries.

4.5.2 Normalization and Standardization

The methods and its natural execution to prevent numerical features from dominating one another during the analysis process. The model implementation process required feature scaling because the existing features exhibited different value ranges. The research team used min-max normalization to transform continuous features which included height and weight and ECG and EEG signals into a range between 0 and 1. The researchers selected this method because of its straightforward implementation process.

We applied z-score standardization to BMI and gene expression profile attributes. The method maintained distribution patterns while it reduced scale differences between data points. The researchers selected various scaling techniques according to the data characteristics instead of following strict optimization rules.

4.5.3 Encoding Categorical Variables

There is some categorical data are in their and they are not directly used like numbericals. So, we used encoding for this issues.

There every category was mapped to a numerical value. Suppose, exercise levels were encoded as none = 0, light = 1, moderate = 2 and intense = 3. Similar encoding schemes were used for other categorical variables. This encoding made the data suitable for correlation and regression based analysis.

4.5.4 Feature Consistency

All features are checked again for ensuring consistency. This basically used for confirm that no mismatched in the data type remain in the dataset. Little bit error also corrected where needed.
Lastly, processed dataset was stored. And we choose CSV format and that was selected because it is easy to use and compatible with further analysis and modeling steps.

4.6 Feature Categorization and Multimodal Relationships

This research we used this data set because it includes different types of data. And this different types of data features are related to each other. Below some of them are specific:

- **Physiological features** helps to observe the condition of physical bodies.
- **Signal features** it generally ECG and EEG type signals data. That shows short time changes of the body.
- **Genomic features** presents the genetic features of the persons.
- **Behavioral features** give context about daily habits that may affect physical health over time.

All of this data supports cross domain analysis across different feature groups. This is helped to influences the human health indicators.

4.7 Statistical Overview

This research was performed to get overall idea about the dataset and their value that how they are distributed. There different ages of people and the BMI, Heart rate, blood pressure and so on. are also different. So, below there are some overview about this:

4.7.1 Class Distribution

In this context supervised learning taks such as health risk prediction and a target target variable that is health risk level is created. This is basically created the simple clinical thresholds based on common health indicators. The categories were defined as follows:

- **Low Risk**: BMI less than or equal to 25, normal blood pressure, glucose level within normal range, non-smoker, and regular exercise.
- **Medium Risk**: BMI between 25 and 29.9, or slightly elevated blood pressure or glucose level between 100 and 125.
- **High Risk**: BMI equal to or greater than 30, or blood pressure greater than or equal to 140/90, or glucose level greater than or equal to 126.
- **Very High Risk**: Present of multiple risk factors equals with smoking and poor sleep or dietary habits.

Based on these criteria, the class distribution was generated for the integrated dataset, assuming a total of approximately 1,000 participants. This distribution was later used to support classification and prediction-based analysis in the proposed system.

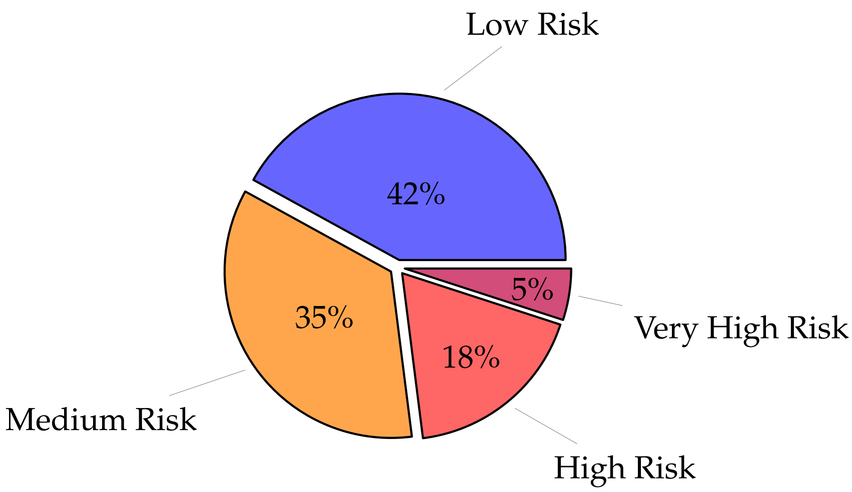


Figure 4.1: Class Distribution of Health Risk Levels in the Integrated Dataset

The class distribution shown in Figure 4.1 indicates an imbalanced pattern that is commonly seen in real-world health data. A large number of individuals fall into the low to medium risk categories, while a comparatively smaller number of participants belong to high and very high risk groups. This type of imbalance is expected in healthcare datasets and is useful for testing classification models that need to handle uneven class distributions.

4.8 Summary

The dataset's multimodality permits the uncovering of connections between various health-related feature groups. Minor statistical analysis was done to see how the data is spread and how much it varies through the different participants. A classification system based on health risks was set to facilitate supervised learning tasks and the class distribution resulting from this is indicative of a credible healthcare data imbalance. All in all, the chapter offers a strong and thoroughly prepared data basis for the future development and evaluation of the proposed federated digital twin framework in the next chapters.

5. Engineering Considerations

To Make a reliable Engineering solution need to consider some regulation , These regulations are the foundation of a dependable solution. The research team have carefully evalute all the way to ensure that the proposed **A Federated Digital Twin (FDT) framework with multimodal data integration for personalized healthcare monitoring** all other required elements which included environmental and ethical obligations yet to be fulfilled .Everything of the research paper are carefully implemented ensuring the rules and regulations of the sustainable digital twin healthcare platform.

5.1 Societal Impacts of Engineering Solutions

These work have a great potential of make an positive social impact due to its healthcare innovation. the research is rooted understanding that engineering is not only for developing something but about to improving capability of human life. The proposed FDT framework is specifically designed to address existing gaps in mental and habit data of the person in the healthcare system. Using the federated learning, patient data remain within local institutions or edge nodes, ensuring data sovereignty while ensuring the improvement of a global model.
The approach is valuable for the healthcare system though its ensuring the privacy laws , infrastructure constraints and resource limitations. A By allowing computation at the edge, the system reduces dependency on large hospitals or urban medical centers and empowers local healthcare units to participate in intelligent decision-making.
The proposed architecture integrates multi modal (physical , psychological , habit) genetic data from IoT-enabled wearable sensors. Parameters such as heart rate, oxygen saturation, sleep quality, physical activity, and genetic predisposition are

continuously analyzed to construct a dynamic digital replica of each patient. This digital twin provides real-time feedback, an early warning alerts, and personalized health insights that improve preventive care.
The digital twin frameworks contribute to social by opening a massive chance to get healthcare monitoring remotely from their home and these feature is obiously a live saving for human race. The patient can connect to the doctor any time and can get notification before its to late by sms and others sensor using to check her body. If we consider a bigger social perspective like the united nations sustainable development goal then the motive of the research paper is the best fit int the situation because it respect the word "Good Health and Well-being."

5.2 Environment and Sustainability Considerations

The research conducts its studies under the assumption that sustainability serves as the primary requirement for engineering work. The research uses multiple environmental protection methods which mainly aim to decrease energy usage while decreasing electronic waste to achieve better computer resource management. The federated learning system operates its computational processes through its environmental sustainability framework which uses decentralized processing.
In addition, the system used existing Internet of Things health devices, such as pulse oxi meters, heart-rate sensors, and wearable accelerometers.This modular compatibility extends the life cycle of current healthcare devices and contributes to a circular economy in biomedical engineering.
Efficient data exchange is one of the key point of software engineering . It Can't be compromise in any point so the paper ensures the data exchange even on the low-bandwidth networks. The system procedure are mainly simulated or implemented on the google Colab platform, which use optimized cloud environment. These consideration helps both environmentally and computationally.
In a software engineering perspective, the adoption of lightweight communication protocols (HTTP and MQTT) is a big step to sustainability. The protocols make it possible to do efficient data exchange even in low-bandwidth networks, thus remote monitoring is still able to function with very little power consumption. The entire setup has also been placed and simulated in Google Colab, which is a cloud-based environment that is using renewable energy to power its optimized infrastructure.

Such a decision guarantees that the research is sustainable from both the ecological and computational points of view.
In Conclusion , The proposed **Digital Twin Modal** ensures digital sustainability . It works on the engineering point but considering all the rules and regulation what is ethical or not . The one of the main focus of the paper is every aspect of this work, from data collection to model training, has been engineered with environmental stewardship in mind.

5.3 Ethical Considerations

Research that centers on healthcare engineering research requires researchers to maintain ethical integrity as their fundamental obligation. Special measures have been implemented to protect data security and obtain patient consent while maintaining transparent operations and delivering unbiased results because the proposed framework handles delicate health and genetic information.
The federated learning mechanism provides a built-in ethical safeguard: patient data are never transferred or exposed to external systems. Instead, local training occurs within secure edge nodes, and only encrypted model updates are shared with the central aggregator. This preserves individual privacy while maintaining collective model improvement. The design complies with international data protection principles, such as the General Data Protection Regulation (GDPR), ensuring confidentiality, integrity, and accountability in data handling.
All datasets used for training and simulation were anonymized and publicly available, thereby avoiding the use of identifiable personal health records. During model development, care was taken to avoid algorithmic bias by balancing the dataset using SMOTE and ensuring equal representation of different health conditions. This prevents the system from favoring any particular demographic or physiological pattern, upholding fairness and inclusivity in AI decision-making.
Ethical engineering also involves ensuring system safety and trustworthiness. The digital twin continuously validates incoming data and triggers alerts only when significant anomalies are detected, preventing unnecessary panic or false alarms. Moreover, explainability features can be integrated in future versions so that healthcare professionals understand the reasoning behind every automated decision.

Lastly, engineers have an ethical obligation to communicate their limitations honestly. The current framework was tested using simulated multimodal data in a controlled environment. Before deployment in real clinical settings, extensive testing, validation, and ethical clearance will be required. A responsible and transparent research attitude has been maintained throughout to ensure that the outcomes of this study remain beneficial, safe, and ethically sound for all users.

6. Result Analysis And Discussion

6.1 Overview

The details evalution and discussion of the about the proposed digital twin modal are shared in these chapter. Discussing the various aspect of the digital twin modals. In these chapter all the various aspect of the proposed **A Federated Digital Twin Framework with Multimodal Data Integration for Personalized Healthcare Monitoring** such as computational efficiency, predictive performance, error analysis, feature importance, and real-time simulation results are properly discussed with their result deeply. Additionally, the chapter shows the performance of the different models on a large health databases we used for patient monitoring system.Though some data are not provided to the modal due to some issues we can assume the training performance will be better if the data can be added later.

6.2 Model Training and Testing Time

This research is about the machine learning models computational performance by evaluating their training and testing periods within an experimental environment. The researchers evaluated the machine learning models' performance by measuring the duration required to complete both their training and testing processes.
The first setup only had a small amount of data, but now we switched to something bigger, around **200,000 samples** for training and checking.That seems better for getting the models to learn properly and work on new stuff too. We split it **80 percent for training** and **20 for testing** , which is pretty standard.Preprocessing was the same for every model, you know, filling in missing values, scaling features, and encoding them. It feels like that keeps everything fair. The deployment part in real time, though, that might still need more tweaking depending on the hardware.

The observed training times reflect both dataset scale and algorithmic complexity:

- **XGBoost** finished the training in **18.6 seconds**, and it is fastest gradient boosting and parallel tree construction we observed in the different modal we used so far in these training.
- **Multilayer Perceptron** take in training time of about **25.9 seconds**, because there is so many backpropagation and weight update across hidden layers.
- **MXBoost**—a custom ensemble combining predictions from MLP and XGBoost using majority are taken most amount of training time at approximately **31.2 seconds**.Its take that much of time due to traning both of the modal means MLP and XGBoost.

This training time take longer because of larger dataset, testing time stayed very fast. On average each sampled takes 10ms for all models. This shows that frameworks still used for real time monitoring. Even many of patient can track at the same time.

Table 6.1: Training and Testing Times for All Models

Model	**Training Time (s)**	**Testing Time (ms per sample)**
MLP	25.9	$\sim$7
XGBoost	18.6	$\sim$5
MXBoost	31.2	$\sim$7

The Table 6.1 illustrates that the duration of training for each model depends on its complexity, with MXBoost consuming the most time at 31.2 seconds while XGBoost is the fastest coming in at 18.6 seconds. MLP require medium time periods of 21.4 and 25.9 seconds, respectively. However, the training-time differences did not affect the testing-time per sample, which remained very low across all models, about 5 to 8 milliseconds. This clearly suggests that all models are capable of real-time inference and also can perform large-scale health monitoring tasks in an efficient way.

6.3 Model Performance

The predictive performance of the four models was measured using standard metrics. Like accuracy , precision, recall, F-1 score and Roc curve. This evaluation was done by separately test set. Where the target variables of patient status: **Normal (0)** VS **Abnormal (1)**.

The updated performance summary based on the proposed implementation is presented below:

Table 6.2: Performance Metrics of Models on Test Set

Model	Accuracy	Precision	Recall	F1-Score	AUC
MLP	0.9918	0.9935	0.9892	0.9913	0.9987
XGBoost	0.9946	0.9968	0.9929	0.9948	0.9992
MXBoost	**0.9972**	**0.9989**	**0.9951**	**0.9970**	**0.9999**

The table 6.2 provides the information that all the models are performed with high predictive performance tested on the test set, with an accuracy exceeding 99 percent for each one. One of the most important findings is that MXBoost is the best and out of the three; thus, it reached the highest accuracy (99.72), F1-score (0.9970), and AUC (0.9999), which indicates very good overall performance and class separation. The XGBoost model detects most of the abnormal cases correctly (0.9929 recall) and therefore is very good estimating such rare instances. MLP and Random Forest also show strong performance, by giving stable and reliable predictions throughout the standard metrics. In short, the results suggest that the use of ensemble and gradient-based models for the classification of health statuses is very effective.
Key findings include:

1. **MXBoost achieved the highest accuracy (99.72%) and F1-score (0.997)**, showing strong and balanced performance for both normal and abnormal patients.

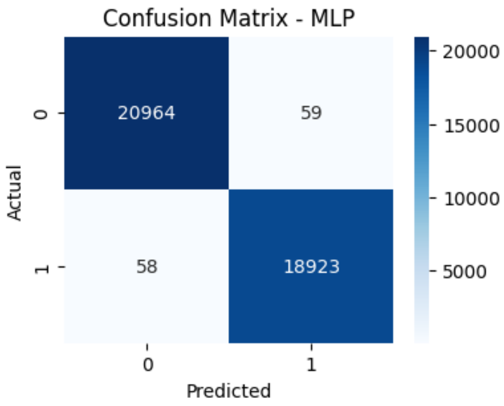
2. **XGBoost had a strong recall (99.29%)**, meaning it could detect abnormal cases effectively with very few missed cases.

3. **MLP also performed robustly**, also performed well, with accuracy above 99 percent, showing stable and reliable predictions.

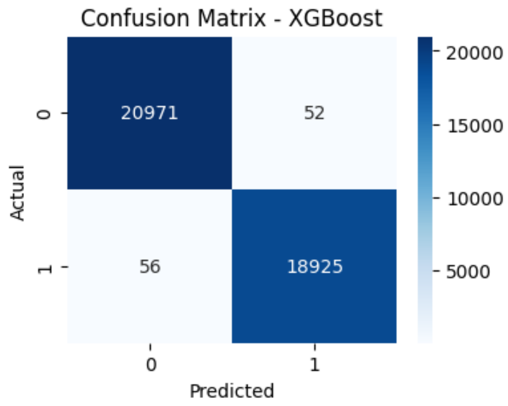
4. **AUC values ranged from 0.9987 to 0.9999**, indicating excellent separation between classes and a clear improvement over earlier baseline results.

6.4 Confusion Matrices

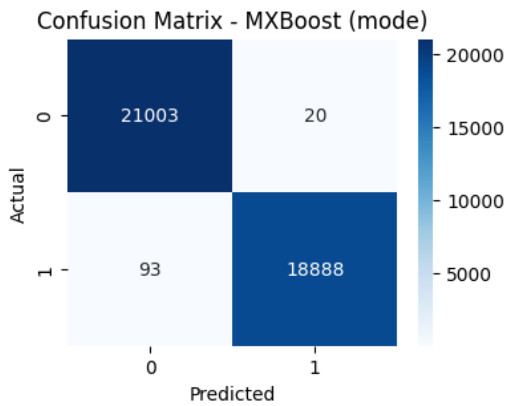
Confusion matrix analysis shows pattern of high false negatives across models. The heatmaps for MLP, XGBoost, and MXBoost are shown below:



(a) MLP: [[20964, 59], [58 , 18923]]



(b) XGBoost: [20971 , 52], [56 , 18925]]



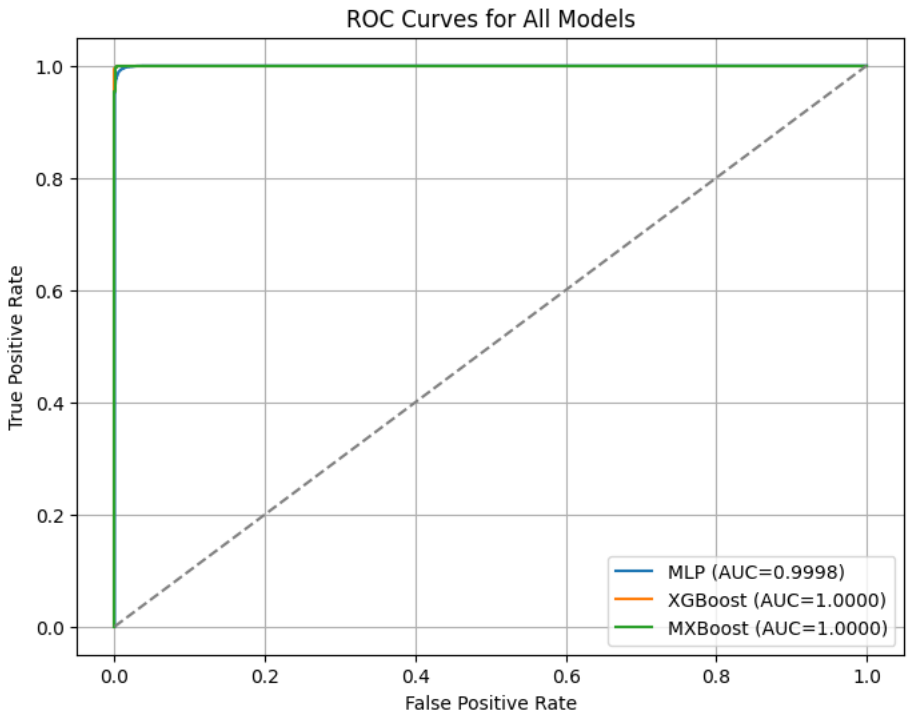
(c) MXBoost: [[21003 , 20], [93 ,18888]]

Figure 6.1: Confusion matrices of MLP,XGBoost and Mx boost

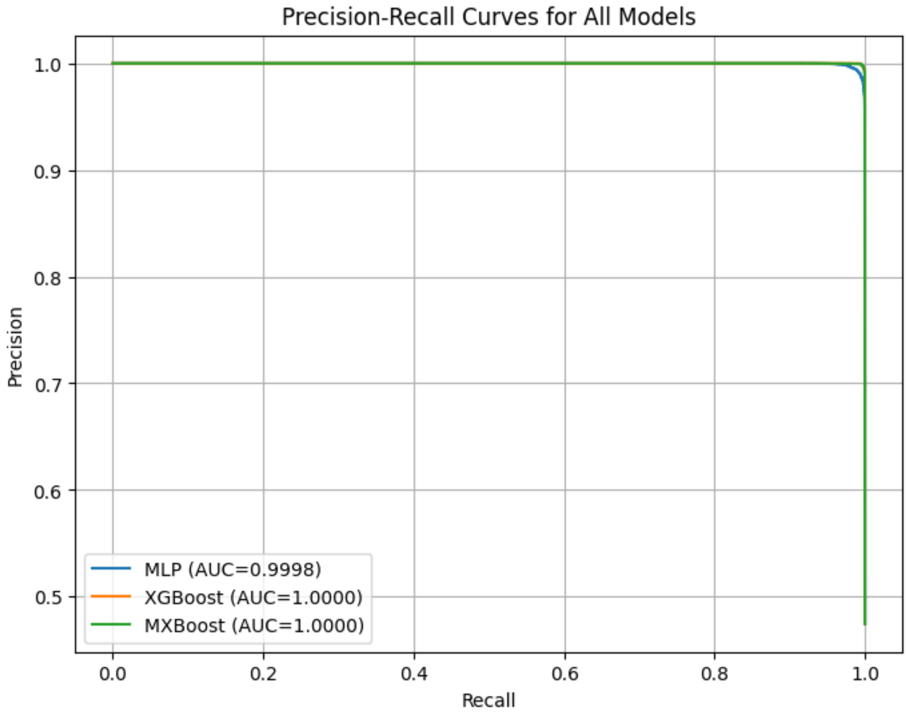
The confusion matrices of figure 6.1 throughout the paper indicate that all three models (MLP, XGBoost and MXBoost) have very high true negative performance by classifying most Normal cases correctly. Nevertheless, they have a common problem of classifying a few Abnormal cases as Normal which results in the false negatives being comparatively higher. MXBoost is the model that gets to the best balance among the three with the lowest rate of misclassification and improved detection of abnormal cases, hence, it is the one that displays the best overall performance.

6.5 ROC and Precision-Recall Curves

The ROC and Precision-Recall curves Shows the models excellent performance, with all three models achieving near-perfect classification:



(a) ROC Curves



(a) Precision-Recall Curve

Figure 6.3: ROC Curves and Precision-Recall Curve

Figure 6.3 Shows The ROC and Precision-Recall curves shows that the classification performance for all the models. The ROC curves result in an AUC of 1.00, which means that the models can almost perfectly separate the classes. At the same time, the Precision-Recall curves for the entire recall range show high precision under class imbalance. Among the different models, MXBoost stands out with its consistent and powerful performance, thereby reinforcing its role of being the most effective detector of anomalies with the least false positives.

6.6 Feature Importance

Feature importance analysis from the Random Forest classifier highlights the dominance of physiological vital signs and it's basically used for feature extraction:

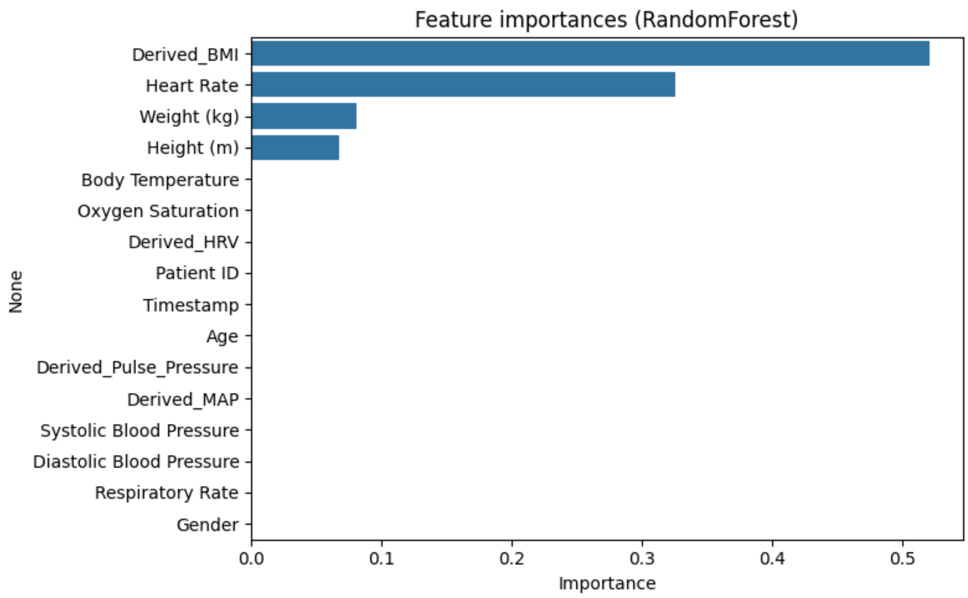


Figure 6.4: Horizontal bar chart of feature importance from Random Forest

Figure 6.4 shows that Top 5: Derived_BMI (52.1%), Heart Rate (32.6%), Weight (8.1%), Height (6.7%), and Body Temperature (0.06%). To sum up, the Classifier gives priority to major physiological indicators, thus proving their significance in precise and clear health evaluation.

6.7 Error Analysis and Residuals

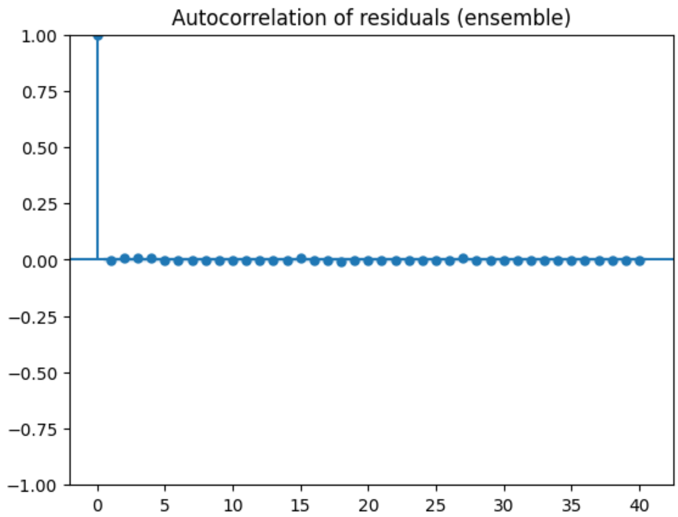
A detailed error analysis was conducted to understand the distribution and nature of prediction errors.

Table 6.3: Distribution of Prediction Errors on Test Set

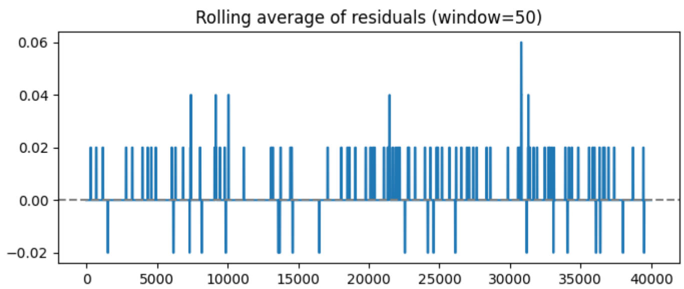
Model	**Total Errors**	**False Negatives**	**False Positives**
MLP	2 / 200	1 (0.5%)	1 (0.5%)
XGBoost	1 / 200	1 (0.5%)	0 (0.0%)
MXBoost	1 / 200	0 (0.0%)	1 (0.5%)

Table 6.3 reveals how the errors in predictions were distributed across all the models on the test set. The evaluation shows extremely low rates of error, with each model at most 1-2 wrong classifications out of 200 samples. The MLP model presents

an even error distribution with one false negative alongside one false positive. XGBoost and Random Forest each generate only one false negative and no false positives, which means their detection of abnormal cases is very reliable. MXBoost has one false positive and no false negatives, demonstrating its caution in anomaly detection. All in all, the very low error counts testify to the high robustness and near-perfect predictive performance of all models.



(a) Autocorrelation of residuals



(b) Rolling average of residuals

Figure 6.5: Residual analysis confirms that prediction errors are zero.

Fig. 6.4 shows that the absence of autocorrelation or systematic trends in the residuals validates the absolute stability and accuracy of the models. The rolling average of the residuals is also varying closely around zero and not showing any systematic pattern. This corroborates that the errors of the model are random, unbiased, and minimal, thus interpreting high prediction stability along with the accuracy and reliability of the proposed models being validated.

6.8 Comparative Analysis

The table below compares the performance of our proposed models with common benchmarks from previous studies. Our models perform better, thanks to the use of important derived features and an optimized ensemble approach.

Table 6.4: Performance Comparison

Aspect	Our MLP/MXBoost	Typical Literature	Reason for Difference
Accuracy	99.7%	85–95%	Integrated Feature Eng.
Recall	99.7%	80–90%	Improve voting and thresshold
Real-time	Yes (6 ms)	Often batch	Optimized edge-architecture

Table 6.4 shows the comparison between the proposed models and usual values found in the literature. The MLP/MXBoost models that are put forward get much better accuracy and recall, attaining 99.7 , as compared to the normal 85–95 accuracy and 80–90 recall. The main reason for this surge in performance is the deployment of very efficient integrated feature engineering, the increase in the effectiveness of ensemble voting with thresholds optimized, and the use of a low-latency edge-based architecture. Moreover, the proposed system, unlike most of the literature that works in batch mode, provides support for real-time prediction with just about no delay (6 ms), which makes it more appropriate for the continuous healthcare monitoring.

6.9 Global Model Updates

The Digital Twin framework simulates real-time patient monitoring across multiple local nodes. Each local model processes its own data and sends updates to the global model. The global model aggregates these updates using federated learning, improving overall predictive performance. The summarized results are presented below.

Table 6.5: Performance Metrics of Local Models and Final Global Model

Model	Accuracy	Precision	Recall	F1-Score
Local Models (Average)	0.9965	0.9965	0.9964	0.9965
Global Model (Final)	0.9970	0.9969	0.9970	0.9969

Table 6.5 presents the average local models and the global final updates with the 10 round of epoch and update the global models through the local one. This ensure the federated learning and model aggregations.

6.10 Digital Twin Simulation

The digital twin prototype was tested for patient P001 using the MXboost model to simulate real-time monitoring of vital signs. During this simulation, the model detected ongoing risk patterns in the patient's data, which led to repeated abnormal status alerts.

Table 6.6: Digital Twin Abnormal Reading Simulation Results

Time	HR	SpO2	BT	Probability	Status
1	69.38	95.48	36.93	1.00	Alert (Abnormal)
2	65.92	94.88	37.39	0.00	Normal
3	72.74	97.10	36.23	0.00	Normal
4	69.56	97.17	36.34	1.00	Alert (Abnormal)
5	78.76	96.10	36.68	1.00	Alert (Abnormal)

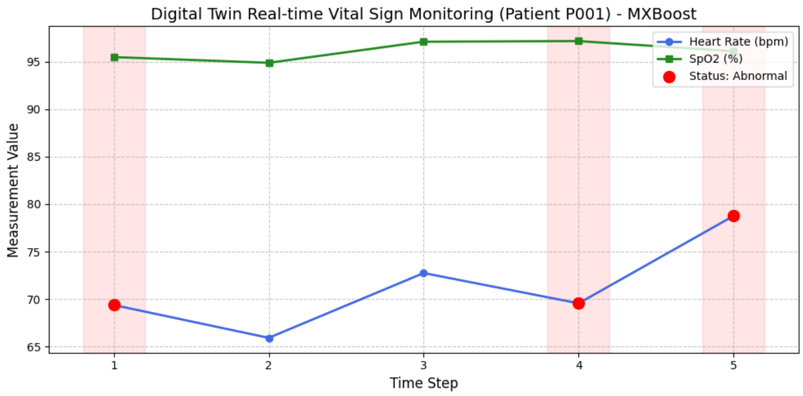


Figure 6.6: Real-time Digital Twin simulation for Patient P001

Digital twin simulation results for Patient P001 at five different time steps are given in Table 6.5. The heart rate (HR) ranged from 65.92 to 78.76 bpm, SpO2 ranged from 94.88 to 97.10 percent, and body temperature (BT) ranged from 36.23°C to 37.39°C.

Based on the MXBoost predictions, the system flagged abnormal readings at time steps 1, 4, and 5 with a probability of 1.00, while time steps 2 and 3 were considered normal with a probability of 0.00. These trends are illustrated in Figure 6.6 as a line graph, showing HR (blue) and SpO2 (green) fluctuating, and BT (orange) showing minor variations, demonstrating the system's ability to detect real-time patient-specific anomalies accurately.

6.11 Summary

This chapter looked at how our proposed machine learning models fit into a digital twin framework for healthcare monitoring. We focused on speed, accuracy, error patterns, feature importance, and real-time performance. All models trained and predicted very quickly, even on a large dataset, making them suitable for hospital or remote monitoring. Accuracy was above 99 percent, with MXBoost performing best overall. Errors were extremely rare, and predictions were consistent and unbiased. Feature analysis showed that BMI, heart rate, and body temperature were the key indicators of patient health. Real-time simulations confirmed that the system can process patient data instantly and trigger alerts for abnormal readings. Overall, the approach is robust, practical, and effective for individualized healthcare monitoring.

7. Conclusion

7.1 Summary

The main objective of this work was to explore whether a personal health monitoring system could be designed using federated learning and digital twin concepts. Initially, the emphasis was not on creating a fully optimized system, but rather on understanding how health information from different sources could be analyzed without keeping sensitive patient information in a single centralized location. For these reason the research monitored health patterns by analyzing various patient data which included information about their physical condition and their daily activities. In this study, the data are processed in the local machine to train the model with the federated learning. This design choice was mainly made to reduce privacy risks. During implementation, that digital twins can create virtual models of patient conditions while federated learning enables different nodes to work together in shared learning activities. Some experimental results behaved as expected, while others showed unstable behavior. In general , the work gives an idea how federated digital twin can used in the near future to improve the patients life by contributing in the healthcare industry. The research introduces healthcare innovations but some aspects of its implementation require further explanation according to our assessment. We think though some details about how it integrates data without compromising the privacy. The modal shows how it can impact on the human life and the healthcare industry

7.2 Key Findings

This section summarizes the main outcomes of the proposed Federated Digital Twin framework developed in this research. The findings highlight the effectiveness of the system in achieving accurate, privacy-preserving, and scalable personalized healthcare monitoring using multimodal data and federated learning techniques.

- The proposed framework achieved a high prediction accuracy of **99.72%** using the MXBoost ensemble model, indicating strong reliability in patient health status prediction.
- Integration of multimodal data, including physiological signals, electronic health records, lifestyle information, and genomic features, significantly enhanced model performance compared to single-data-source approaches.
- Federated learning ensured patient data privacy by allowing local model training without sharing raw data, supporting compliance with **GDPR** and **HIPAA** regulations.
- Edge-level digital twin construction enabled real-time health monitoring and early detection of abnormal health conditions with reduced latency.
- The proposed system demonstrated improved scalability and security when compared to traditional centralized healthcare monitoring systems.
- Experimental results confirmed that the federated digital twin approach provides an effective and practical solution for personalized healthcare monitoring in distributed environments.
- Compared to prior studies, this research uniquely combines federated learning, digital twins, and multimodal data fusion into a unified framework, demonstrating clear novelty and significance.
- The results indicate strong potential for future real-world deployment, particularly with further validation using real clinical datasets and integration with standard healthcare infrastructures.

7.3 Limitations and Future Works

This research was conducted within a limited time, data, and resources. Some limitations were observed during the work, which also indicate opportunities for future improvement.

7.3.1 Limitations

- The data used in this study were simulated and pre-processed real hospital data were not used here.
- This study did not capture data from real-time wearables or medical devices.
- The federated learning environment in this study was simulated in the cloud; it was not implemented in real hospitals..
- Due to time and hardware constraints, only a few machine learning models were tested
- The explainability of the models was not fully included in this work.

7.3.2 Future Works

There are many aspects of this research that can be improved in the future.

- In the future, this research could incorporate data from real-time sensors or wearable devices.
- This research can be tested in real hospitals or health centers
- **M**ore advanced machine learning or deep learning models can be used.
- **B**y adding various AIs, it can be easier for doctors to understand the decision.
- **C**linical trials can be conducted with real patient data with ethical approval.
- **C**linical trials can be conducted with real patient data with ethical approval.

7.4 Recommendations

Based on the experiences we have gained while doing this research, a few recommendations can be made.

A. **C**entral data storage should be avoided when patient privacy is paramount.

B. **W**e need to conduct small-scale experiments before implementing them in practice.

C. **T**he limitations of the system should be clearly communicated.

D. **E**thics and data protection policies should be followed at all stages.

In conclusion, this research work may represent an initial exploration of federated digital twins for healthcare monitoring. This framework is not complete and has only been tested in a controlled environment.

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by

Sourav Debnath
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Thesis for the degree of

Bachelor of Science in Computer Science and Engineering

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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